Getting Started with r4c

r4c package

2026-06-30

Overview

`r4c` supports two operational rangeland tasks:

1. Classifying range condition from six-factor field sheets.
2. Estimating dry fodder production and livestock units from cutting and weighing (CW), comparative yield (CY), and double sampling (DS) data.

The package accepts Excel workbooks and plain R data frames. The included example workbook is available after installation:

```r
path <- system.file("extdata", "r4c_data.xlsx", package = "r4c")
basename(path)
#> [1] "r4c_data.xlsx"
```

Installation

Install from GitHub:

```r
install.packages("remotes")
remotes::install_github("babaknaimi/r4c")
```

For a local checkout, run:

```r
install.packages("devtools")
devtools::install(".")
```

Then load the package:

```r
library(r4c)
```

Excel Workbook Structure

The example workbook uses these sheets:

```r
readxl::excel_sheets(path)
#> [1] "sfd1"   "sfd2"   "cyd"    "dsd"    "re_cwd"
```

The range-condition sheets use the field-template structure:

- `SP`: species or special row label.
- `PC`: palatability class for species rows.
- `C1`, `C2`, …: canopy or cover measurements.
- `S1`, `S2`, …: seedling or supplementary observations.
- Special rows: `LFP`, `SGP`, `TCP`, and `BSP`.

The production sheets are:

- `cyd`: comparative-yield calibration data with `Xri`, `Ygi`, and `Ydi`.
- `dsd`: double-sampling calibration data with `Xej`, `Ygj`, and `Ydj`.
- `re_cwd`: field data with CW observations and CY/DS predictor columns.

Range Condition

In the example workbook, `sfd1` and `sfd2` are treated as two consecutive years. The helper below reads the field-template layout, separates `PC` from the cover columns, and classifies each year.

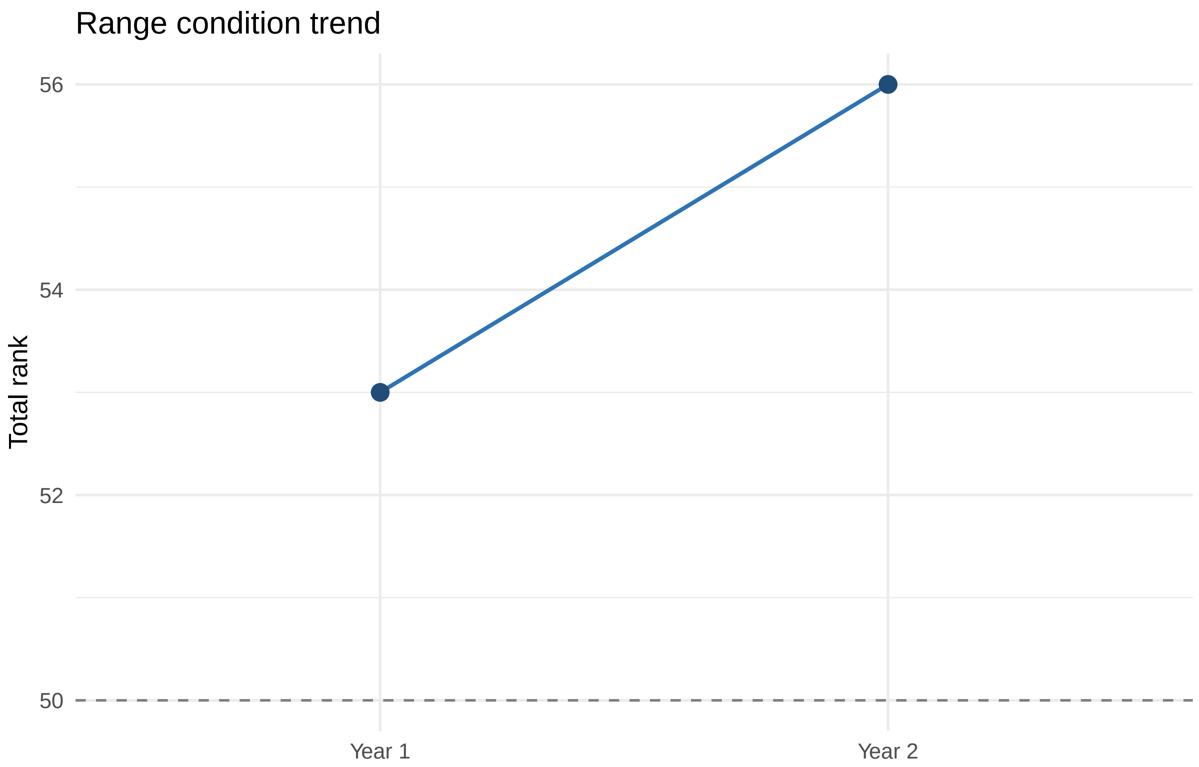
```r
condition <- classify_sf_template_years(
  path,
  sheets = c("sfd1", "sfd2"),
  year_names = c("Year 1", "Year 2"),
  score_profile = "template_rank"
)

condition$summary
#>     year total_score condition_class
#> 1 Year 1          53            Fair
#> 2 Year 2          56            Fair
```

The template-rank profile is intentionally explicit. The attached materials did not include the official six-factor scoring table, so `r4c` exposes the default profile through `score_max`, `count_reference`, and `rank_fun`. Replace those arguments if your agency uses a different official rubric.

```
condition$details[["Year 1"]]$factor_scores
#>           factor value_pct max_score   score
#> 1    composition  67.26519        50 33.6326
#> 2     production  34.80000        15  5.2200
#> 3         canopy  34.80000        15  5.2200
#> 4         litter   1.55000        10  3.1000
#> 5         vigour   0.90000         5  0.9000
#> 6 soil_protection  37.25000        15  5.5875
```

```
ggplot(condition$summary, aes(year, total_score, group = 1)) +
  geom_line(linewidth = 0.8, color = "#2E74B5") +
  geom_point(size = 3, color = "#1F4D78") +
  geom_hline(yintercept = 50, linetype = 2, color = "gray50") +
  labs(
    x = NULL,
    y = "Total rank",
    title = "Range condition trend"
  ) +
  theme_minimal()
```



Fodder Production and Livestock Units

Use `estimate_fodder_production()` when you want CW, CY, and DS summarized from the `re_cwd` sheet. The default livestock-unit calculation uses:

- allowable use: 0.60
- range area: 700 ha
- animal-unit weight: 60 kg
- daily intake: 2% of body weight
- grazing period: 6 months
- days per month: 30

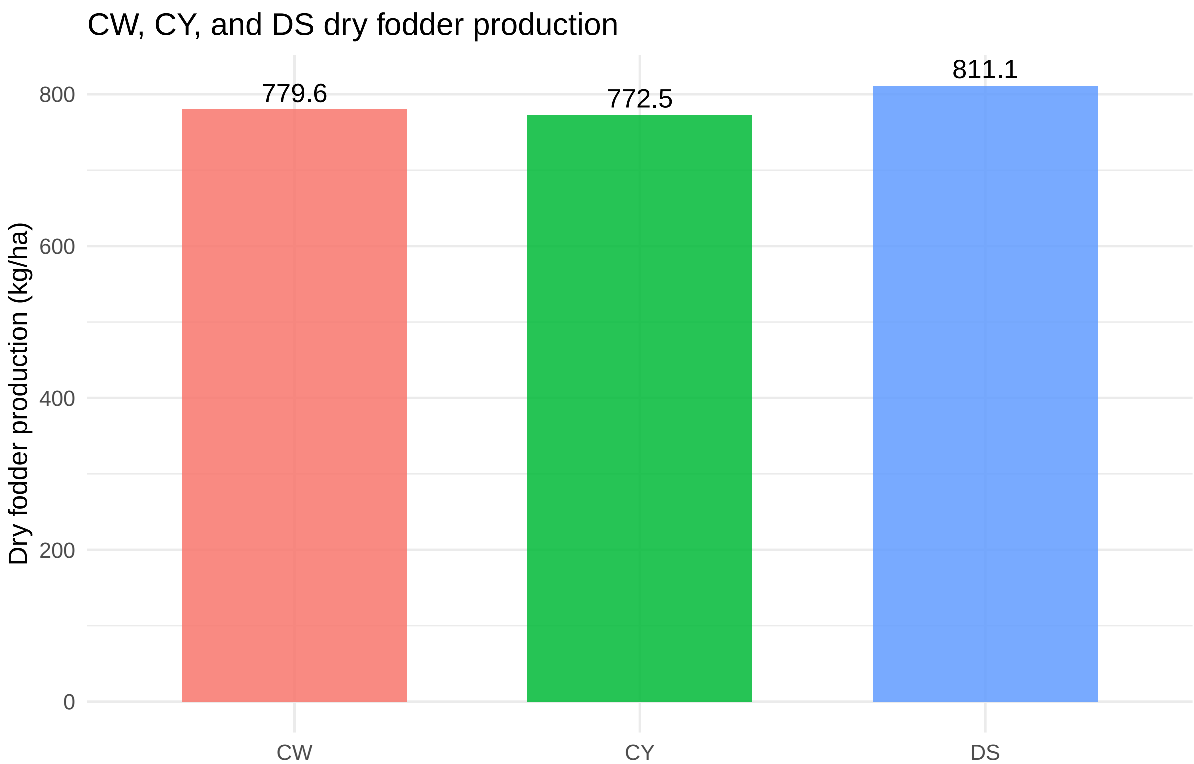
```
production <- estimate_fodder_production(path)

production_summary <- production$summary
production_summary$dry_fodder_kg_ha <- round(production_summary$dry_fodder_kg_ha, 1)
production_summary$total_available_fodder_kg <- round(production_summary$total_available_fodd
production_summary$livestock_units <- round(production_summary$livestock_units)
production_summary
```

```
#>   method dry_fodder_kg_ha total_available_fodder_kg livestock_units
#> 1     CW            779.6                  327413.3            1516
#> 2     CY            772.5                  324441.9            1502
#> 3     DS            811.1                  340673.0            1577
```

For this workbook, the livestock-unit checks round to 1516, 1502, and 1577 for CW, CY, and DS. The continuous CY regression gives 772.5 kg/ha. A legacy spreadsheet that rounds intermediate CY predictions can display about 773.1 kg/ha, so confirm which convention your organization wants before locking a reporting template.

```
ggplot(production_summary, aes(method, dry_fodder_kg_ha, fill = method)) +
  geom_col(width = 0.65, alpha = 0.85) +
  geom_text(aes(label = dry_fodder_kg_ha), vjust = -0.4) +
  labs(
    x = NULL,
    y = "Dry fodder production (kg/ha)",
    title = "CW, CY, and DS dry fodder production"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



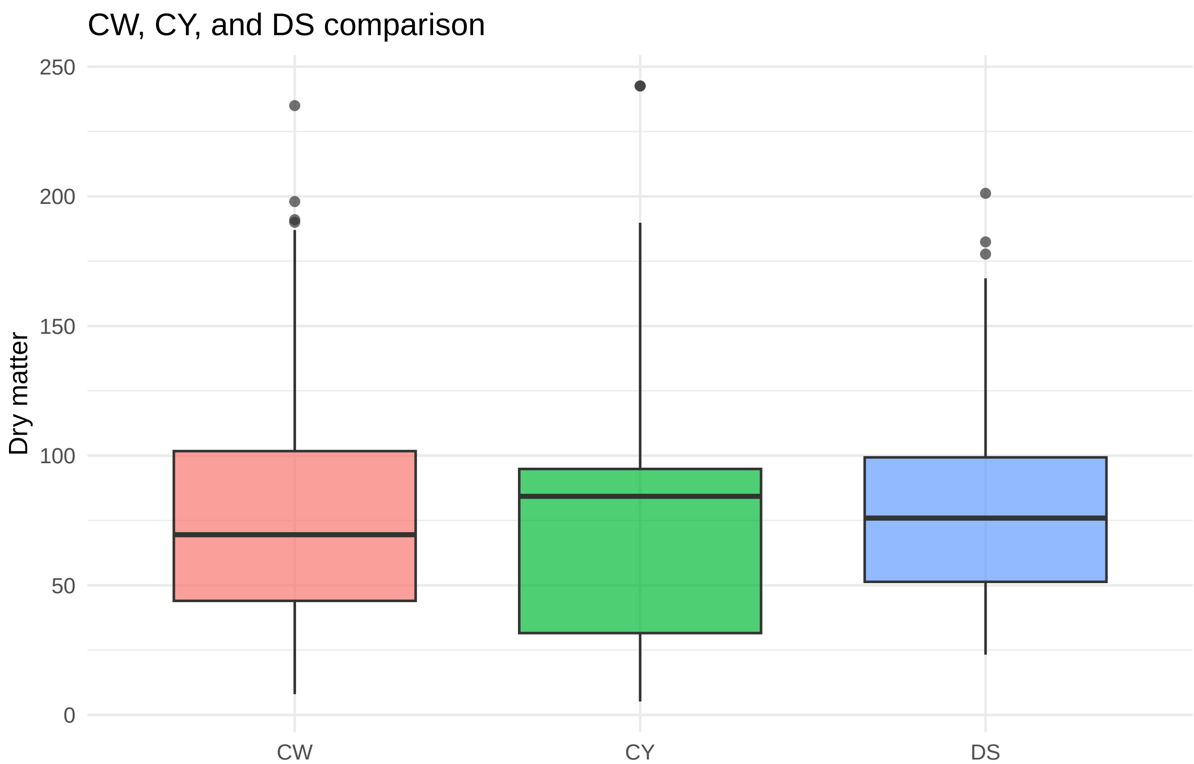
Optional Validation

CW is not required for routine CY or DS estimation, but it is useful for method comparison. Use `validate_methods()` to build a statistical comparison and `plot_validation()` for diagnostic plots.

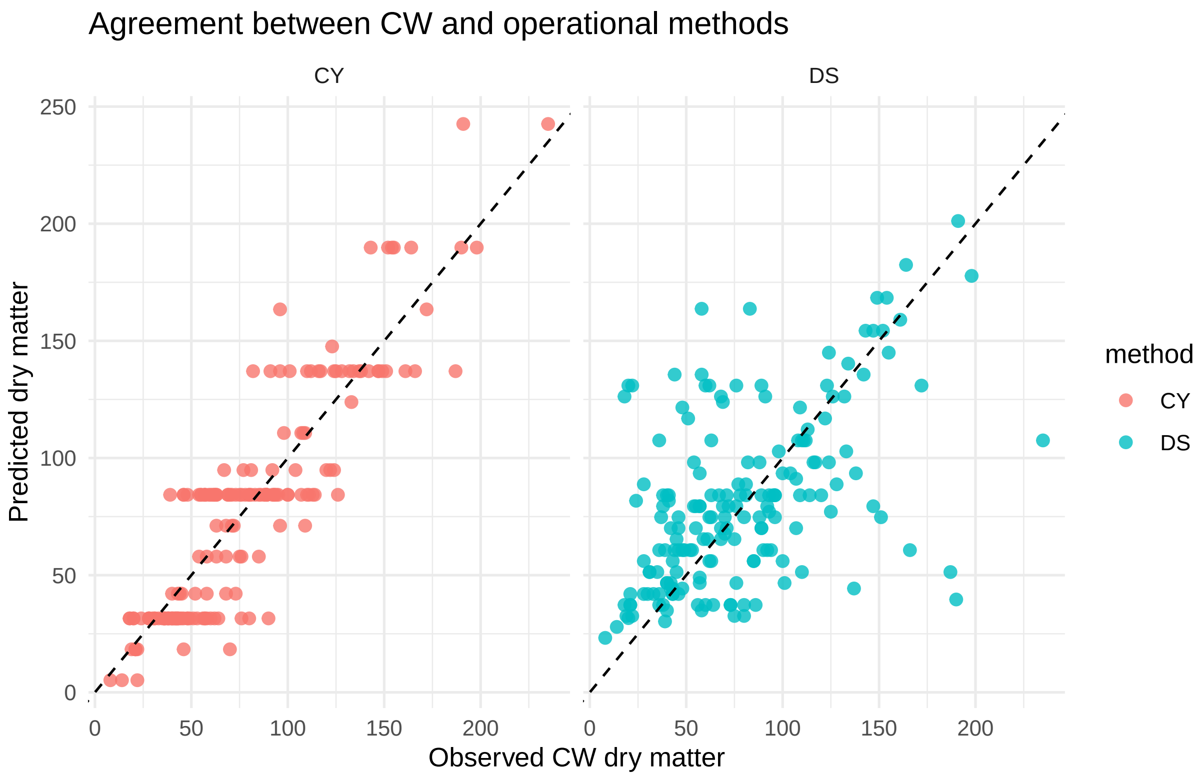
```
cyd <- read_cyd_excel(path)
dsd <- read_dsd_excel(path)
cwd <- read_cwd_excel(path)

cc <- carrying_capacity(cyd, dsd)
validation <- validate_methods(cwd, cc, test = "kruskal")
validation
#> <r4c_validation_result>
#>
#>  Kruskal-Wallis rank sum test
#>
#> data:  Yd by Method
#> Kruskal-Wallis chi-squared = 2.1861, df = 2, p-value = 0.3352
```

```
plot_validation(validation, type = "boxplot")
```



```r
plot_validation(validation, type = "agreement")
```



Working with Your Own Data

Use a workbook with the same sheet names:

```r
my_path <- "path/to/my_r4c_workbook.xlsx"

classify_sf_template_years(
  my_path,
  sheets = c("sfd1", "sfd2"),
  year_names = c("Year 1", "Year 2"),
  score_profile = "template_rank"
)

estimate_fodder_production(
  my_path,
  allowable_use = 0.60,
```

```r
  range_area_ha = 700,
  au_weight_kg = 60,
  grazing_months = 6
)
```

For data already loaded into R, pass data frames directly:

```r
cyd <- read_cyd_excel(my_path)
dsd <- read_dsd_excel(my_path)
cwd <- read_cwd_excel(my_path)

estimate_fodder_production(cyd, dsd, cwd)
```

Reporting Checklist

Before reporting results, check:

1. The SF sheets represent the intended unit: replicated sheets for one year should be merged; sheets from different years should be classified separately.
2. The `PC` palatability classes match the local field guide.
3. The six-factor scoring profile matches your official rubric.
4. The production parameters match the management question: area, allowable use, animal-unit weight, grazing months, and daily intake.
5. CY/DS validation is only interpreted when the CW comparison data are relevant and collected under the same field conditions.